

Artificial Intelligence & Machine Learning

Autonomous Multi-Agent Negotiation for Crisis Resource Allocation

Background:

During natural disasters, critical city infrastructure such as power grids, emergency services, hospitals, and water systems often operate independently. This lack of coordination leads to inefficient resource usage and can trigger cascading failures. Human decision-makers are unable to negotiate and reallocate resources quickly enough in high-pressure, rapidly evolving scenarios.

Problem Statement:

There is a need for an intelligent, decentralized system where autonomous agents representing different utilities can negotiate resource allocation in real time. The system must ensure efficient distribution of limited and fluctuating power supply while strictly adhering to system constraints, avoiding inconsistencies, and minimizing overall damage and loss of life during crisis situations.

Expected Outcomes:

- A real-time simulation demonstrating coordinated negotiation between multiple agents under constrained conditions.
- Efficient and stable allocation of power resources with minimized service disruptions and failure states.
- Intelligent decision-making by agents that balances priority, fairness, and system-wide impact.
- A scalable framework capable of handling dynamic disaster scenarios over extended periods.

Real-Time, Multimodal Deepfake Attribution and Detection

Background:

Deepfake technology has rapidly evolved from simple visual manipulation to sophisticated multimodal generation, combining face synthesis with highly realistic voice cloning. These deepfakes can now be generated in real time during live video interactions, making them significantly harder to detect. Existing detection systems are often too slow or limited to a single modality, and they lack the ability to identify the underlying generative model responsible for the fake content.

Problem Statement:

There is a need for a lightweight, real-time system capable of analyzing synchronized audio and video streams to accurately detect deepfakes under strict latency constraints. Beyond detection, the system should perform attribution by identifying the likely type or family of generative model used, while maintaining high accuracy and operating efficiently on standard consumer hardware.

Expected Outcomes:

- A real-time multimodal detection system with low latency suitable for live or streaming scenarios.
- Accurate identification of deepfake content using both visual and audio cues.
- Capability to attribute detected deepfakes to probable generative model families.
- An optimized solution that balances performance, accuracy, and computational efficiency for practical deployment.

Privacy-Preserving Synthetic Healthcare Data Generator**Background:**

Medical research and AI-driven healthcare innovations rely heavily on access to Electronic Health Records (EHRs). However, strict privacy regulations such as HIPAA and GDPR severely limit data sharing due to the sensitive nature of patient information. Traditional anonymization techniques are insufficient, as they can still lead to re-identification risks. This creates a critical need for solutions that enable data usability while ensuring strong privacy guarantees.

Problem Statement:

There is a need to develop a generative AI system capable of producing high-quality synthetic healthcare data that preserves the statistical properties and predictive utility of real patient records. The system must incorporate rigorous privacy-preserving mechanisms, such as Differential Privacy, to ensure that no individual patient's information can be inferred or reconstructed from the generated data.

Expected Outcomes:

- A synthetic dataset that closely matches the statistical distribution and relationships of the original healthcare data.
- Strong privacy guarantees ensuring resistance to re-identification and membership inference attacks.
- Comparable performance of machine learning models trained on synthetic data versus real data.
- A robust and scalable pipeline capable of handling structured records, temporal data, and unstructured clinical text.

Cybersecurity and blockchain

AI-Driven Next-Generation Firewall for Dynamic Threat Detection and Zero Trust Implementation

Background:

Modern enterprise networks face increasingly sophisticated cyber threats, including encrypted attacks, zero-day exploits, and insider breaches. Traditional Next-Generation Firewalls (NGFWs) rely heavily on static rules and signature-based detection, making them ineffective against evolving attack patterns. Additionally, limited integration with Zero Trust principles and inadequate use of AI-driven analytics restrict their ability to respond proactively in dynamic, distributed environments.

Problem Statement:

Current NGFWs lack the intelligence required for advanced threat detection, real-time adaptation, and automated mitigation. They struggle to analyze encrypted traffic efficiently, adapt to emerging Indicators of Compromise (IoCs) and MITRE ATT&CK techniques, and enforce continuous, context-aware security policies. There is a need for an AI-driven firewall system that can dynamically learn, detect, and respond to threats while integrating seamlessly with Zero Trust architectures.

Expected Outcomes:

- A high-performance prototype capable of real-time threat detection and mitigation with minimal latency.
- Adaptive and intelligent security policies that evolve based on emerging threats and behavioral patterns.
- Seamless implementation of Zero Trust principles with continuous authentication and fine-grained access control.
- Scalable deployment across multi-cloud, hybrid, and edge environments with enterprise-grade throughput and compliance support.

The Adaptive "Chameleon" Honeypot Infrastructure

Background:

Traditional honeypots are static and easily identifiable by sophisticated attackers, particularly Advanced Persistent Threats (APTs). Once detected, attackers simply bypass them, rendering these systems ineffective. To counter modern threats, deception systems must be dynamic, adaptive, and capable of convincingly mimicking real-world environments to engage and study attacker behavior.

Problem Statement:

There is a need to develop an intelligent, adaptive honeypot system that can dynamically modify its services, vulnerabilities, and responses based on real-time attacker behavior. The system should convincingly emulate different environments, engage attackers through tailored interactions, and securely capture their actions without exposing the host infrastructure to risk.

Expected Outcomes:

- A dynamic honeypot system capable of adapting its behavior and services in real time based on attack patterns.
- Realistic and engaging deception environments that effectively trap and delay attackers.
- Comprehensive logging and analysis of attacker activities, including payload capture and behavioral insights.
- A secure, sandboxed architecture ensuring complete isolation and protection of the host system.

Secure Data Wiping for Trustworthy IT Asset Recycling

Background:

India is facing a rapidly growing e-waste crisis, with over 1.75 million tonnes generated annually. A significant contributor to this issue is the reluctance of individuals and organizations to dispose of old electronic devices due to concerns about data security. Sensitive personal and corporate data stored on laptops and smartphones creates fear of data breaches if devices are recycled or resold. Existing data sanitization tools are often complex, costly, or lack verifiable proof of erasure. This has resulted in large volumes of unused devices being stockpiled, slowing down efforts toward a circular economy and sustainable e-waste management.

Problem Statement:

There is a need to design a secure, user-friendly, and cross-platform data wiping solution that ensures complete and verifiable erasure of sensitive data across Windows, Linux, and Android devices. The system must provide strong guarantees of data sanitization, support offline usage, and generate tamper-proof certification that can be independently verified, while remaining accessible to non-technical users.

Expected Outcomes:

- A reliable data wiping application capable of securely erasing all user data, including hidden and low-level storage areas.
- Generation of digitally signed, tamper-proof wipe certificates in multiple formats for audit and verification purposes.
- A simple, intuitive interface enabling one-click operation for general users.
- Offline functionality through bootable media for secure and isolated execution.
- A scalable and standards-compliant solution that enhances trust in device recycling and promotes responsible e-waste management.

